Data streaming:

1- Data freshness

```
bruno@MyMacbookPro deel-data-engineering-task % docker-compose exec transactions-db psql -U finance_db_user -d finance_db -c "
SELECT
    MAX(to_timestamp(event_ts_ms / 1000.0))         AS latest_event,
    NOW() - MAX(to_timestamp(event_ts_ms / 1000.0)) AS data_age
FROM analytics.fact_orders;"
        latest_event        |    data_age
----------------------------+-----------------
 2026-10-14 23:38:00.264+00 | 00:00:33.433188
(1 row)
```

2- Data counts in destination (Analytics schema in PostgewSQL database)

```
bruno@MyMacbookPro deel-data-engineering-task % docker-compose exec transactions-db psql -U finance_db_user -d finance_db -c "
SELECT NOW()::time AS queried_at,
       (SELECT COUNT(*) FROM analytics.fact_orders)           AS orders,
       (SELECT COUNT(*) FROM analytics.fact_order_items)      AS order_items,
       (SELECT COUNT(*) FROM analytics.dim_customers)         AS customers,
       (SELECT COUNT(*) FROM analytics.dim_products)          AS products,
       (SELECT COUNT(*) FROM analytics.orders_status_history) AS history;"
   queried_at    | orders | order_items | customers | products | history
-----------------+--------+-------------+-----------+----------+---------
 23:40:05.646959 |    326 |        8476 |        20 |       28 |    1147
(1 row)
```

3- Events and latency

```
bruno@MyMacbookPro deel-data-engineering-task % docker-compose exec transactions-db psql -U finance_db_user -d finance_db -c "
SELECT order_id, status, op, changed_at, ingested_at,
       ingested_at - changed_at AS pipeline_lag
FROM analytics.orders_status_history
ORDER BY ingested_at DESC LIMIT 10;"
 order_id |    status    | op |       changed_at        |      ingested_at       | pipeline_lag
----------+--------------+----+-------------------------+------------------------+--------------
      392 | PENDING      | c  | 2026-10-14 23:43:00.282 | 2026-10-14 23:43:11.66 | 00:00:11.378
      392 | PROCESSING   | u  | 2026-10-14 23:43:00.284 | 2026-10-14 23:43:11.66 | 00:00:11.376
      381 | COMPLETED    | u  | 2026-10-14 23:43:00.276 | 2026-10-14 23:43:11.66 | 00:00:11.384
      383 | PROCESSING   | u  | 2026-10-14 23:43:00.282 | 2026-10-14 23:43:11.66 | 00:00:11.378
      379 | REPROCESSING | u  | 2026-10-14 23:43:00.283 | 2026-10-14 23:43:11.66 | 00:00:11.377
      367 | PROCESSING   | u  | 2026-10-14 23:43:00.283 | 2026-10-14 23:43:11.66 | 00:00:11.377
      389 | REPROCESSING | u  | 2026-10-14 23:43:00.275 | 2026-10-14 23:43:11.66 | 00:00:11.385
      368 | PROCESSING   | u  | 2026-10-14 23:43:00.276 | 2026-10-14 23:43:11.66 | 00:00:11.384
      391 | PENDING      | c  | 2026-10-14 23:43:00.276 | 2026-10-14 23:43:11.66 | 00:00:11.384
      380 | REPROCESSING | u  | 2026-10-14 23:43:00.284 | 2026-10-14 23:43:11.66 | 00:00:11.376
(10 rows)
```

4 – Full Journey of an order:

```
bruno@MyMacbookPro deel-data-engineering-task % docker-compose exec transactions-db psql -U finance_db_user -d finance_db -c "
SELECT status, op, changed_at
FROM analytics.orders_status_history
WHERE order_id = 123 ORDER BY changed_at;"
  status   | op |       changed_at
-----------+----+------------------------
 PENDING   | c  | 2026-10-14 23:32:00.334
 COMPLETED | u  | 2026-10-14 23:32:00.337
(2 rows)
```